

System Architecture Version 1

1. Components

1.1 Client Layer

- **Web Application / Mobile App**
 - User interface for interaction
 - Sends requests to backend services
 - Receives responses and displays data

1.2 API Gateway

- Acts as the single entry point for all client requests
- Handles request routing, authentication, and rate limiting
- Forwards requests to appropriate backend services

1.3 Backend Services

- **Authentication Service**
 - Manages user login, registration, and token issuance
- **Business Logic Service**
 - Processes core application logic
 - Handles data processing and validation
- **Data Service**
 - Manages data storage and retrieval
 - Interfaces with the database

1.4 Data Storage

- **Relational Database (e.g., PostgreSQL)**
 - Stores structured data such as user profiles, transactions, etc.
- **Cache Layer (e.g., Redis)**
 - Stores frequently accessed data for quick retrieval
 - Reduces load on the primary database

1.5 Infrastructure Components

- **Load Balancer**
 - Distributes incoming traffic across multiple backend instances
- **Monitoring & Logging**
 - Tracks system health and logs events for troubleshooting

2. Data Flow

1. **Client Request**

- User interacts via Web/Mobile app
- Sends HTTP request to API Gateway

2. **Request Handling**

- API Gateway authenticates the request (via tokens)
- Routes request to the appropriate backend service

3. **Processing**

- Backend Service processes the request
- Retrieves or updates data in the Data Storage
- Uses Cache Layer to optimize read operations

4. **Response**

- Backend Service sends response back to API Gateway
- API Gateway forwards response to the client

5. **Logging & Monitoring**

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1.4 Data Storage

- **Relational Database (e.g., PostgreSQL)**
 - Stores structured data such as user profiles, transactions, etc.
- **Cache Layer (e.g., Redis)**
 - Stores frequently accessed data for quick retrieval
 - Reduces load on the primary database

1.5 Infrastructure Components

- **Load Balancer**
 - Distributes incoming traffic across multiple backend instances
- **Monitoring & Logging**
 - Tracks system health and logs events for troubleshooting

2. Data Flow

1. **Client Request**

- User interacts via Web/Mobile app

- Sends HTTP request to API Gateway

2. **Request Handling**

- API Gateway authenticates the request (via tokens)
- Routes request to the appropriate backend service

3. **Processing**

- Backend Service processes the request
- Retrieves or updates data in the Data Storage
- Uses Cache Layer to optimize read operations

4. **Response**

- Backend Service sends response back to API Gateway
- API Gateway forwards response to the client

5. **Logging & Monitoring**

- All

System Architecture Version 1

1. Components

1.1 Client Layer

- **Web Application / Mobile App**
 - User interface for interaction
 - Sends requests to backend services
 - Receives responses and displays data

1.2 API Gateway

- Acts as the single entry point for all client requests
- Handles request routing, authentication, and rate limiting
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1.3 Backend Services

- **Authentication Service**
 - Manages user login, registration, and token issuance
- **Business Logic Service**
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